

Geofence Event Processing Challenge

Scenario

You're building a location-based service for a taxi company. The system needs to track vehicles as they move through different geographic zones and detect when they cross zone boundaries based on GPS coordinates received. Assume you will receive real-time GPS location event updates directly from vehicles.

Requirements

Your service should:

1. Accept location events via HTTP endpoint
2. Detect and report when vehicles enter or exit a few simple geographic zones
3. Provide a way to query the current zone status for a vehicle

Please spend no more than 2 hours on this challenge. Use any languages, libraries, or frameworks you deem appropriate. Your solution should demonstrate practical engineering judgment, code quality, and operational awareness.

What we're looking for

We're evaluating your engineering judgment and practical skills, not just your ability to implement algorithms.

Specifically:

- Code quality: Clean, readable, maintainable code with good structure
- Decision making: Thoughtful architectural choices with clear reasoning
- Operational awareness: Appropriate error handling, logging, and monitoring considerations
- Performance thinking: Awareness of efficiency concerns and scalability
- Edge cases: Identification and handling of boundary conditions
- Communication: Clear documentation of assumptions, tradeoffs, and future improvements

Deliverables

Submit your solution as a zip file containing:

1. Complete source code
2. README.md with setup instructions, assumptions, and design decisions

3. Any configuration files needed to run your service
4. Notes on what you would improve given more time

Feel free to use any resources, libraries, or tools at your disposal.